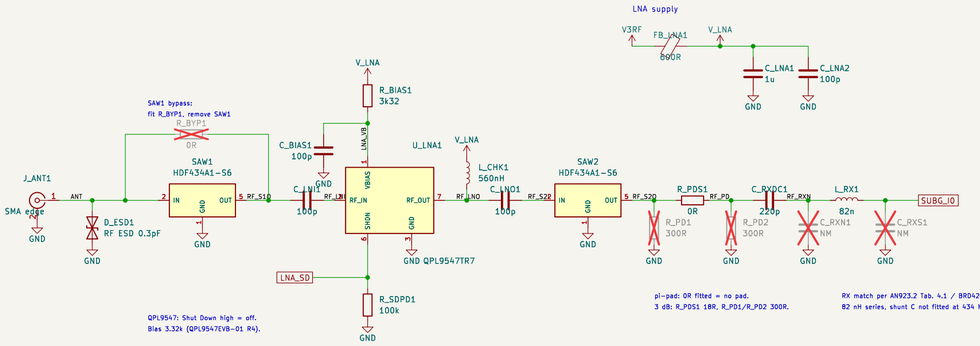
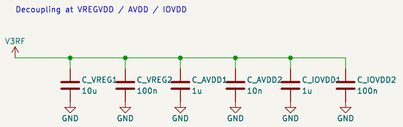
RF front end, 430–440 MHz

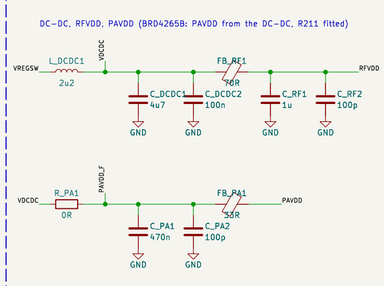
SMA – RF ESD – SAW1 – QPL9547 (–25 dB) – SAW2 – pi-pad option – RX match – FG23 SUBG_IO. Gain to the FG23 –21 dB, system NF ~2.6 dB (0.6 dB with SAW1 bypassed). Shield can over SAW1 … FG23 + TCXO.

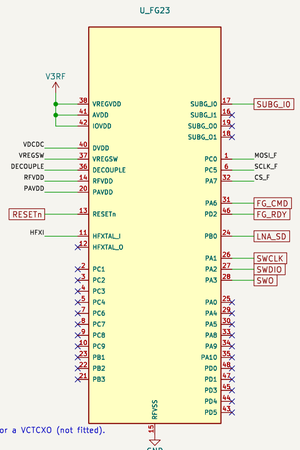


EFR32FG23 radio (RX only)

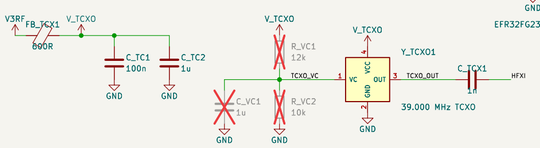
Link pins = ionos firmware: PC00 MOSI, PC05 SCLK, PA07 CS, PA06 CMD (rx), PD02 RDY (driven by the ESP32), PA03 SWO. 22R source termination at the driver. Supply values as fg23-cw-trainer A5 / BRD4265B.

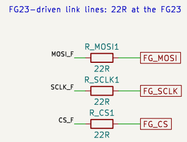


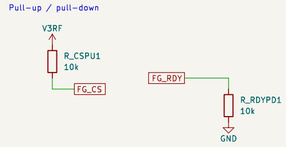


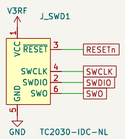


39 MHz TCXO (clipped sine) –> 1 nF –> HFXTAL_I; HFXTAL_O open. Vc parts only for a VCTCXO (not fitted).





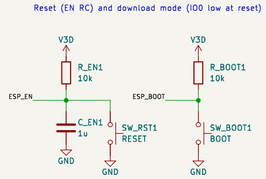


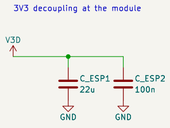


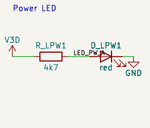
FG23 SWD: Tag-Connect TC2030 (same pinout as fg23-cw-trainer):
the ESP32 also bit-bangs SWD on these nets (firmware update from the phone).

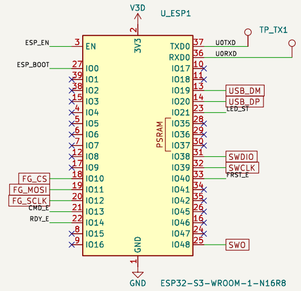
ESP32-S3-WROOM-1-N16R8

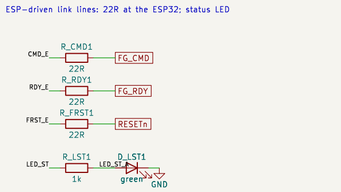
Same module as the ionos-mini DevKit: SPI2 IO_MUX slave for the FG23 (IO10–12), TinyUSB CDC to the phone, SWD bit-bang to the FG23. IO3 / IO45 / IO46 (strapping) and IO35–37 (octal PSRAM) unused.





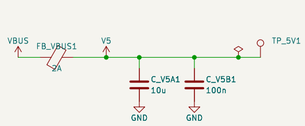


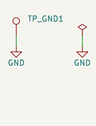


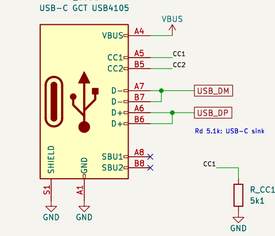


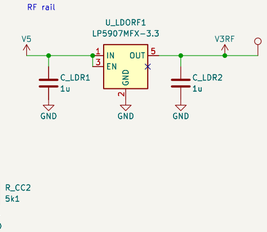
USB-C (OTG device) and supplies

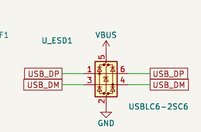
Phone 5 V –> ferrite –> V5 –> AP2112K-3.3 (ESP32-S3, WiFi peaks) and LP5907-3.3 (clean RF rail: FG23, LNA, TCXO). USB 2.0 sink: 5.1k Rd on CC1 / CC2.











Design notes

Spec: claude/SatHunter-FG23-OTG-nyak-spec-2026-11-07 (project) and hardware/fg23-otg/README.md. Parts and nets: generator/design.py (single source of truth).
Checks: check_netlist.py (drawing == design.py), twin_check.py (netlist vs datasheets / firmware / pin map, self-test with injected faults).
To verify before layout: HDF434A1-S6 and QPL9547 land patterns vs the maker drawings; 560 nH choke SRF; TCXO and RF ESD part numbers in the JLC tool.
Layout: 4 layers JLC04161H-7628, RF CPWG ~50 ohm, shield can over SAW1 … FG23 + TCXO, ESP antenna at a board edge with copper keep-out.
In-band spurs to measure on rev A: 440 MHz (ESP 40 MHz x11), 429 MHz (39 MHz x11, SPI 19.5 MHz x22), 432 / 444 MHz (USB 12 / 48 MHz).

CERN-OHL-P-2.0
generated by generator/gen_sch.py from generator/design.py – do not hand-edit
SMA – SAW – QPL9547 – SAW – EFR32FG23 (RX) – ESP32-S3-WROOM-1 – USB-C OTG to the Sath
HA7DCD / Z2 Labs
Sheet: /
File: sathunter_fg23.kicad_sch

Title: SatHunter FG23 OTG – 70 cm satellite receiver

Size: A2	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1